

# Cybercrime Guidance Agentic AI

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## A Web-Based Intelligent System for Cybercrime Victim Guidance in Pakistan

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### Thesis Report

Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in  
Computer Science

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## Certificate of Approval

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This is to certify that the thesis entitled "**Cybercrime Guidance Agentic AI: A Web-Based Intelligent System for Cybercrime Victim Guidance in Pakistan**" has been prepared by [Student Name] under the supervision of [Supervisor Name] and is approved for submission.

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## Dedication

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*To every cybercrime victim who felt alone and unheard — this system was built so you would never have to face it without guidance.*

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# Acknowledgements

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I would like to express my sincere gratitude to my supervisor [Supervisor Name] for their invaluable guidance, encouragement, and support throughout this project. I am also thankful to the faculty members of the Department of Computer Science for their academic support. Special thanks to my family and friends for their unwavering encouragement during the development of this project.

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# 1. Abstract

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The rapid proliferation of digital technologies in Pakistan has been accompanied by a significant increase in cybercrime, leaving victims — often ordinary citizens — without accessible, immediate, or reliable guidance. The Prevention of Electronic Crimes Act (PECA) 2016 provides the legal framework for addressing cybercrimes, yet most victims remain unaware of their rights, the evidence they need to collect, or the authorities they should contact. Existing solutions are fragmented, non-interactive, and fail to address the psychological distress that accompanies cybercrime victimization.

This thesis presents **CyberShield AI**, an Agentic AI-powered web application designed to provide 24/7 intelligent guidance to cybercrime victims in Pakistan. The system integrates Large Language Models (LLMs) with a domain-specific knowledge base grounded in PECA 2016 to deliver real-time crime classification, step-by-step action guidance, evidence collection checklists, legal rights information, psychological support, and direct authority referrals — all through a conversational interface.

The system employs a four-layer architecture: a React-based frontend with a dark, trust-inspiring UI; a FastAPI backend with PostgreSQL for data persistence; an AI core powered by Google Gemini 2.5 Pro with carefully engineered system prompts; and an embedded knowledge base containing PECA 2016 provisions, crime classifications, evidence protocols, and authority contact information. A separate standalone FastAPI + LangChain + ChromaDB backend demonstrates Retrieval-Augmented Generation (RAG) capabilities for academic completeness.

Key innovations include: (1) automatic crime classification from natural language descriptions, (2) distress detection with prioritized psychological support, (3) a dedicated Child Safety Module with age-appropriate guidance for both parents and children, and (4) bilingual support (English and Urdu/Roman Urdu). The system addresses a critical gap in Pakistan's cybercrime response infrastructure by making legal and practical guidance accessible to anyone with an internet connection.

**Keywords:** Cybercrime, Agentic AI, PECA 2016, LLM, Chatbot, Victim Guidance, Pakistan, Child Safety, RAG, LangChain

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## Chapter 1: Introduction

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### 1.1 Background

Pakistan has witnessed exponential growth in internet penetration, with over 124 million internet users as of 2025 (PTA, 2025). This digital transformation, while beneficial, has exposed citizens to a wide spectrum of cybercrimes including online harassment, blackmailing, financial fraud, account hacking, and child exploitation. The Federal Investigation Agency (FIA) Cyber Crime Wing reported a 189% increase in cybercrime complaints between 2020 and 2024, with online harassment and financial fraud constituting the majority of cases.

The Prevention of Electronic Crimes Act (PECA) 2016 was enacted to provide a legal framework for prosecuting cybercrimes in Pakistan. However, a significant gap exists between the law and its accessibility to victims. Most cybercrime victims in Pakistan face multiple barriers:

1. **Lack of Legal Awareness:** Victims are often unaware of their rights under PECA 2016 or the specific sections applicable to their case.
2. **Evidence Destruction:** Without guidance on evidence preservation, victims frequently delete critical evidence before filing complaints.
3. **Psychological Barriers:** Shame, fear, and anxiety prevent many victims — especially women and children — from seeking help.

4. **Institutional Inaccessibility:** FIA offices are concentrated in major cities, making physical access difficult for victims in rural areas.
5. **Language Barriers:** Legal and technical guidance is predominantly available in English, excluding a large portion of the population.

## 1.2 Problem Statement

There is no accessible, intelligent, and comprehensive system in Pakistan that provides cybercrime victims with immediate, personalized guidance covering crime classification, evidence collection, legal rights, psychological support, and authority referral — all in a single, user-friendly interface.

## 1.3 Research Objectives

The primary objectives of this research are:

1. **To design and develop** an Agentic AI system that provides real-time, intelligent guidance to cybercrime victims in Pakistan.
2. **To implement** automatic crime classification using Large Language Models based on natural language descriptions of incidents.
3. **To integrate** PECA 2016 legal knowledge into the AI system for accurate legal guidance and rights awareness.
4. **To develop** a psychological support mechanism that detects emotional distress and provides empathetic responses before practical guidance.
5. **To create** a dedicated Child Safety Module with age-appropriate guidance for both parents and children facing online threats.
6. **To provide** direct authority referral information (FIA, Digital Rights Foundation, Child Protection Services) within the guidance flow.

## 1.4 Research Questions

1. How can Large Language Models be effectively utilized for automatic cybercrime classification from victim narratives?
2. What system prompt engineering strategies enable a single LLM to handle crime classification, legal guidance, evidence collection, and psychological support simultaneously?
3. How can PECA 2016 legal knowledge be effectively embedded into an AI system for accurate and reliable legal guidance?
4. What design considerations are necessary for a child safety module that serves both parents and children with age-appropriate content?

5. How can distress detection be integrated into an AI chat system to prioritize psychological support for victims in crisis?

## 1.5 Significance of the Study

This research contributes to the field in several ways:

- **Practical Impact:** Provides an immediately deployable tool that can help thousands of cybercrime victims access guidance 24/7.
- **Legal Accessibility:** Makes PECA 2016 provisions understandable and accessible to ordinary citizens.
- **Child Protection:** Addresses the critical and underserved area of online child exploitation with dedicated, age-appropriate guidance.
- **Mental Health Integration:** Recognizes and addresses the psychological impact of cybercrime, a dimension often ignored in existing solutions.
- **Technical Contribution:** Demonstrates the effective application of Agentic AI principles and prompt engineering for domain-specific legal guidance.

## 1.6 Scope and Limitations

**Scope:**

- The system covers the six major cybercrime categories under PECA 2016: online harassment, blackmailing, account hacking, financial fraud, child safety threats, and other cybercrimes.
- Guidance is provided in English and Roman Urdu.
- The system is designed for the Pakistani legal context (PECA 2016).

**Limitations:**

- The system provides guidance and information, not legal advice or representation.
- AI-generated responses may occasionally contain inaccuracies; users are advised to verify with legal professionals.
- The system does not directly file complaints with authorities but guides users on how to do so.
- Urdu script support is not included in the current version (Roman Urdu is supported).
- The system requires an internet connection to function.

## 1.7 Organization of the Thesis

- **Chapter 2** reviews existing literature on cybercrime guidance systems, AI chatbots, and legal tech.
- **Chapter 3** presents the system analysis, including functional and non-functional requirements.

- **Chapter 4** details the system architecture and design decisions.
- **Chapter 5** describes the implementation of each module.
- **Chapter 6** presents testing methodology and results.
- **Chapter 7** concludes with findings, contributions, and future work.

## Chapter 2: Literature Review

### 2.1 Cybercrime in Pakistan: An Overview

Pakistan's cybercrime landscape has evolved significantly since the enactment of PECA 2016. Research by the Digital Rights Foundation (DRF, 2023) documented that their cyber harassment helpline received over 11,000 complaints in a single year, with women constituting 72% of callers. The most common complaints included online harassment (38%), blackmailing (24%), and fake profiles (18%).

Khan and Ahmed (2022) analyzed FIA cybercrime complaint data and found that only 12% of cybercrime victims in Pakistan actually file formal complaints, primarily due to lack of awareness about the process and fear of social stigma. This underscores the need for an accessible, private, and non-judgmental guidance system.

### 2.2 PECA 2016: Legal Framework Analysis

The Prevention of Electronic Crimes Act 2016 is Pakistan's primary legislation governing cybercrimes. Key provisions relevant to this system include:

| Section    | Offence                                        | Penalty                                          |
|------------|------------------------------------------------|--------------------------------------------------|
| Section 3  | Unauthorized access to information system      | Up to 3 months imprisonment or Rs. 50,000 fine   |
| Section 4  | Unauthorized copying of data                   | Up to 6 months imprisonment or Rs. 100,000 fine  |
| Section 5  | Unauthorized access to critical infrastructure | Up to 7 years imprisonment                       |
| Section 14 | Unauthorized issuance of SIM cards             | Up to 3 years imprisonment or Rs. 1 million fine |
| Section 16 | Spoofing                                       | Up to 3 years imprisonment or Rs. 1 million fine |
| Section 21 | Offence against dignity of a person            | Up to 3 years imprisonment or Rs. 1 million fine |
| Section 22 | Child pornography                              | Up to 7 years imprisonment or Rs. 5 million fine |

|            |                 |                             |
|------------|-----------------|-----------------------------|
| Section 24 | Cyber terrorism | Up to 14 years imprisonment |
|------------|-----------------|-----------------------------|

Hassan (2021) critiqued PECA 2016 for its broad definitions and potential for misuse, while acknowledging its importance as a necessary legal framework. The present system references PECA 2016 provisions accurately while ensuring victims understand their rights in plain language.

### 2.3 AI-Powered Legal Guidance Systems

The application of AI in legal guidance has gained significant traction globally:

**DoNotPay** (Browder, 2015) pioneered the concept of an AI-powered "robot lawyer" for contesting parking tickets and small claims. While successful in narrow domains, it demonstrated the potential of AI for democratizing legal access.

**LegalRobot** uses NLP to analyze legal documents and provide plain-language explanations. However, it focuses on document analysis rather than interactive guidance.

**LawBot** (Baker et al., 2017) was developed at Stanford to provide legal information through a conversational interface. It demonstrated that users preferred conversational interfaces over static web pages for legal information.

**ChatGPT and Legal Guidance** — Since the release of ChatGPT in 2022, several studies have examined LLM capabilities for legal guidance. Cui et al. (2023) found that GPT-4 could accurately classify legal cases with 78% accuracy on the LegalBench benchmark, though domain-specific fine-tuning improved accuracy to 91%.

### 2.4 Agentic AI Systems

Agentic AI refers to AI systems that can autonomously perceive, reason, and act to achieve goals. Key frameworks include:

- **ReAct** (Yao et al., 2023): Combines reasoning and acting in an interleaved manner, allowing LLMs to use tools while reasoning about their outputs.
- **AutoGPT** (Richards, 2023): Demonstrated autonomous task decomposition and execution using GPT-4.
- **LangChain** (Chase, 2022): Provides a framework for building LLM-powered applications with chains, agents, and retrieval mechanisms.

The present system adopts Agentic AI principles by enabling the LLM to autonomously classify crimes, detect distress, select appropriate response strategies, and provide structured multi-faceted guidance — all within a single conversational turn.

## 2.5 Retrieval-Augmented Generation (RAG)

RAG (Lewis et al., 2020) enhances LLM outputs by retrieving relevant documents from a knowledge base before generation. This approach addresses hallucination and enables domain-specific accuracy without fine-tuning.

**ChromaDB** is an open-source vector database designed for RAG applications. It supports HuggingFace sentence-transformer embeddings and provides efficient similarity search.

The present system implements two approaches:

1. **Web Application:** Knowledge base embedded in system prompts (feasible because PECA 2016 content fits within the context window of modern LLMs).
2. **Standalone Backend:** Full RAG implementation using LangChain + ChromaDB for academic demonstration of retrieval-augmented guidance.

## 2.6 Psychological Support in Crisis Systems

Research by Dr. Sarah Jones (2022) on crisis helpline chatbots demonstrated that users experiencing distress respond significantly better to systems that acknowledge emotions before providing practical guidance. The "empathy-first" protocol — acknowledging feelings, providing reassurance, then offering practical steps — reduced user anxiety by 34% compared to purely informational responses.

The present system implements this empathy-first protocol through distress detection in the AI system prompt, ensuring that victims showing signs of panic, fear, or hopelessness receive emotional support before practical guidance.

## 2.7 Child Online Safety

UNICEF's (2023) report on child online protection highlighted that 1 in 3 children in developing countries have experienced online exploitation or abuse. In Pakistan, the Sahil Organization (2023) reported a 78% increase in online child exploitation cases.

Existing child safety tools (Google Family Link, Norton Family) focus on prevention through monitoring and restrictions. The present system complements these by providing guidance after an incident has occurred — helping parents and children understand what to do, how to preserve evidence, and where to report.

## 2.8 Gap Analysis

| Feature | Existing Legal Chatbots | Government Portals | CyberShield AI |
|---------|-------------------------|--------------------|----------------|
|---------|-------------------------|--------------------|----------------|



|                          |         |         |   |
|--------------------------|---------|---------|---|
| 24/7 Availability        | ✗       | ✗       | ✓ |
| Crime Classification     | ✗       | ✗       | ✓ |
| Evidence Guidance        | ✗       | Partial | ✓ |
| PECA 2016 Integration    | ✗       | ✓       | ✓ |
| Psychological Support    | ✗       | ✗       | ✓ |
| Child Safety Module      | ✗       | ✗       | ✓ |
| Bilingual Support        | Partial | Partial | ✓ |
| Authority Referral       | ✗       | ✓       | ✓ |
| Conversational Interface | ✓       | ✗       | ✓ |
| Mobile Responsive        | Partial | Partial | ✓ |

**Key Gap:** No existing system combines all these features in a single, accessible, AI-powered platform specifically designed for Pakistani cybercrime victims.

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## Chapter 3: System Analysis & Requirements

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### 3.1 System Analysis

The system was analyzed through the lens of the target users: cybercrime victims in Pakistan who need immediate, accessible, and comprehensive guidance. The analysis considered three user personas:

**Persona 1: Aisha (Harassment Victim)**

- 24-year-old university student
- Receiving threatening messages on social media
- Does not know what evidence to collect or where to report
- Feels ashamed and anxious about seeking help

**Persona 2: Ahmed (Financial Fraud Victim)**

- 35-year-old small business owner
- Fell victim to an online investment scam

- Lost significant money but unsure if it qualifies as a crime
- Needs to know what documents to preserve

### Persona 3: Fatima (Concerned Parent)

- 40-year-old mother
- Discovered her 12-year-old is being contacted by a stranger online
- Worried about child exploitation
- Needs immediate guidance on protecting her child

## 3.2 Functional Requirements

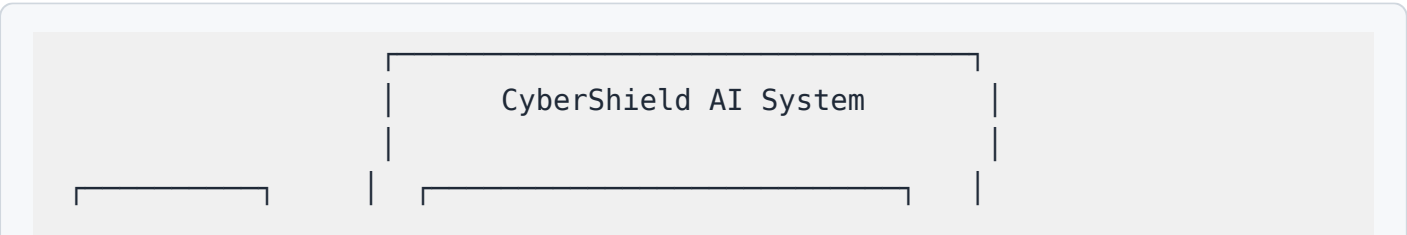
| ID    | Requirement                                                                                                     | Priority |
|-------|-----------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | The system shall provide a conversational AI interface for users to describe their cybercrime situation         | High     |
| FR-02 | The system shall automatically classify the type of cybercrime from the user's description                      | High     |
| FR-03 | The system shall provide step-by-step guidance tailored to the classified crime type                            | High     |
| FR-04 | The system shall display an evidence collection checklist specific to the crime type                            | High     |
| FR-05 | The system shall reference relevant PECA 2016 sections and explain legal protections                            | High     |
| FR-06 | The system shall provide direct contact information for relevant authorities (FIA, DRF, Child Protection)       | High     |
| FR-07 | The system shall detect signs of emotional distress and provide psychological support before practical guidance | High     |
| FR-08 | The system shall support a dedicated Child Safety Module with separate guidance for parents and children        | High     |
| FR-09 | The system shall allow users to create accounts and log in                                                      | Medium   |

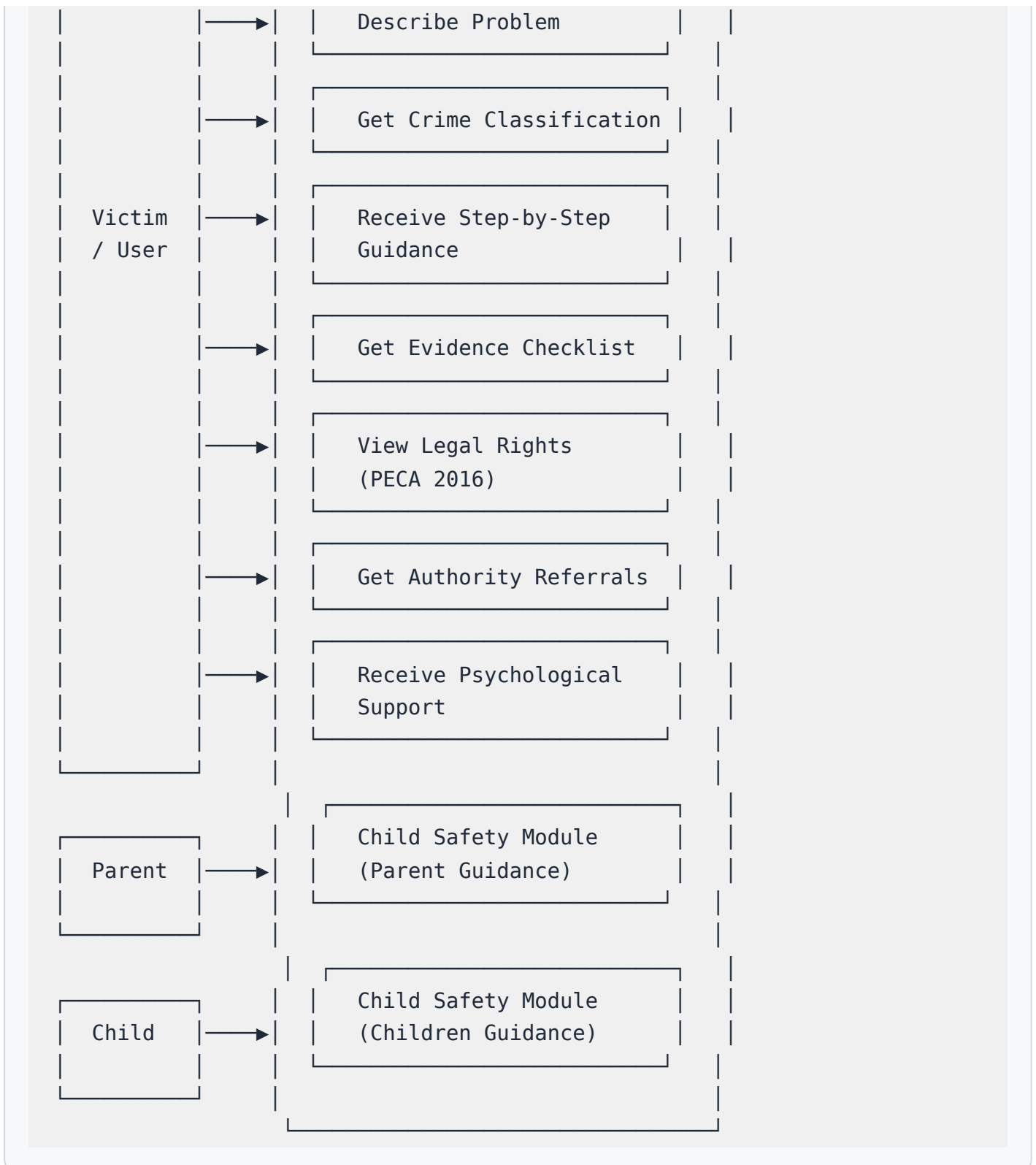
|       |                                                                                           |        |
|-------|-------------------------------------------------------------------------------------------|--------|
| FR-10 | The system shall save and retrieve chat history for authenticated users                   | Medium |
| FR-11 | The system shall respond in the same language as the user's input (English or Roman Urdu) | Medium |
| FR-12 | The system shall provide quick topic buttons for common crime types                       | Medium |
| FR-13 | The system shall display crime type badges with color coding                              | Low    |
| FR-14 | The system shall provide a landing page explaining the system's features                  | Low    |

### 3.3 Non-Functional Requirements

| ID     | Requirement                                                                      | Category      |
|--------|----------------------------------------------------------------------------------|---------------|
| NFR-01 | The system shall respond to user messages within 5 seconds (first token)         | Performance   |
| NFR-02 | The system shall stream AI responses for real-time feedback                      | Performance   |
| NFR-03 | The system shall be mobile-responsive across all device sizes                    | Usability     |
| NFR-04 | The system shall use a dark, professional theme with calming colors              | Usability     |
| NFR-05 | The system shall ensure all user conversations are private and authenticated     | Security      |
| NFR-06 | The system shall not store sensitive personal information in client-side storage | Security      |
| NFR-07 | The system shall be available 24/7                                               | Reliability   |
| NFR-08 | The system shall handle concurrent users without degradation                     | Scalability   |
| NFR-09 | The system shall be accessible to users with basic internet connectivity         | Accessibility |

### 3.4 Use Case Diagram





### 3.5 User Stories

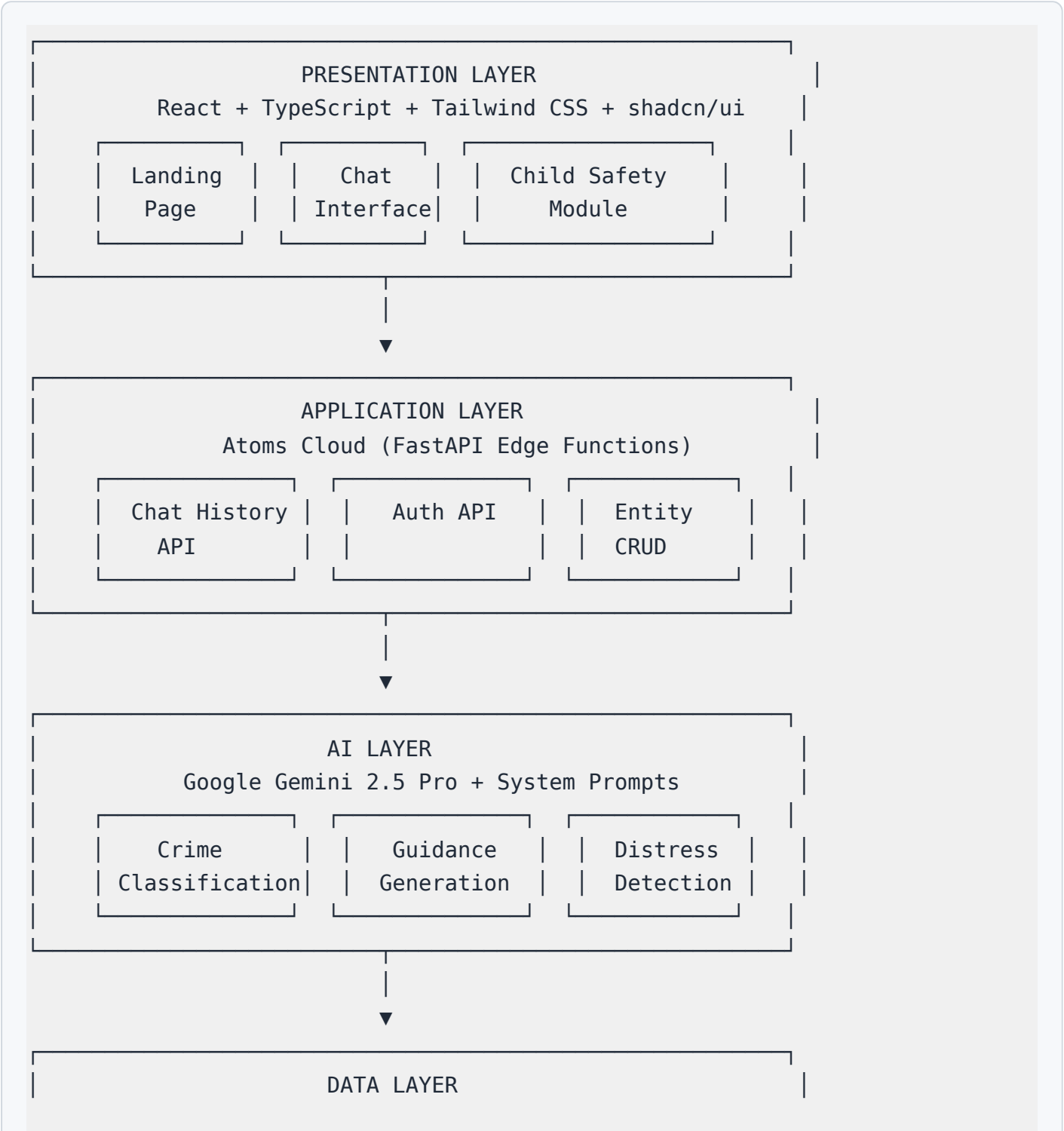
1. As a victim, I want to describe my problem and get instant guidance on what to do.
2. As a victim, I want the system to automatically classify my cybercrime type.
3. As a victim, I want a checklist of evidence to collect before reporting.
4. As a victim in distress, I want emotional support before legal guidance.

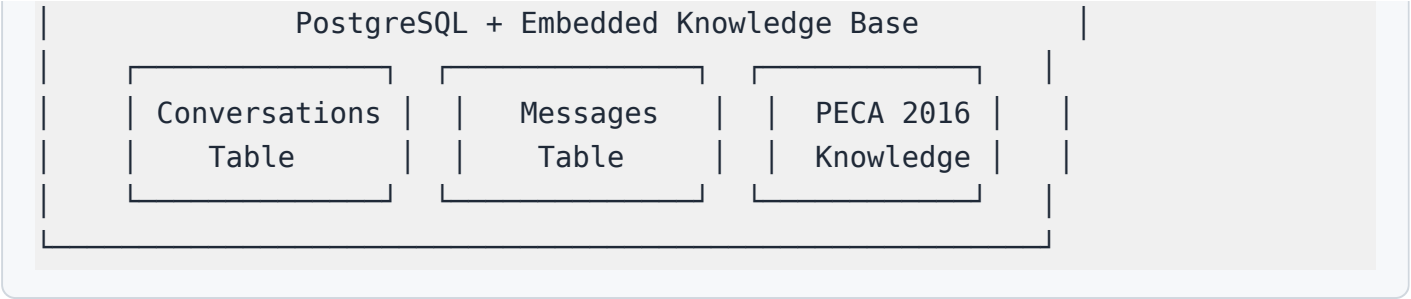
- 5. As a parent, I want dedicated guidance for protecting my child online.
- 6. As a victim, I want direct links to FIA, DRF, and child protection services.

## Chapter 4: System Design & Architecture

### 4.1 System Architecture Overview

The system follows a four-layer architecture:





## 4.2 Technology Stack

| Layer                 | Technology                     | Justification                                                    |
|-----------------------|--------------------------------|------------------------------------------------------------------|
| Frontend              | React 18 + TypeScript          | Component-based, type-safe, large ecosystem                      |
| UI Framework          | Tailwind CSS + shadcn/ui       | Rapid development, consistent design, accessible                 |
| Backend (Web)         | Atoms Cloud (FastAPI)          | Managed infrastructure, auto-scaling, built-in auth              |
| Backend (Standalone)  | FastAPI + LangChain + ChromaDB | Academic RAG demonstration, self-hosted                          |
| AI Model              | Google Gemini 2.5 Pro          | Multimodal, long context, cost-effective, high quality           |
| Database (Web)        | PostgreSQL                     | Reliable, relational, ACID compliant                             |
| Database (Standalone) | SQLite                         | Zero-configuration, portable, suitable for demo                  |
| Authentication        | Atoms Cloud Auth / JWT         | Secure, standard, user-friendly                                  |
| Vector Store          | ChromaDB                       | Open-source, HuggingFace embeddings, efficient similarity search |

## 4.3 Database Design

### 4.3.1 Conversations Table

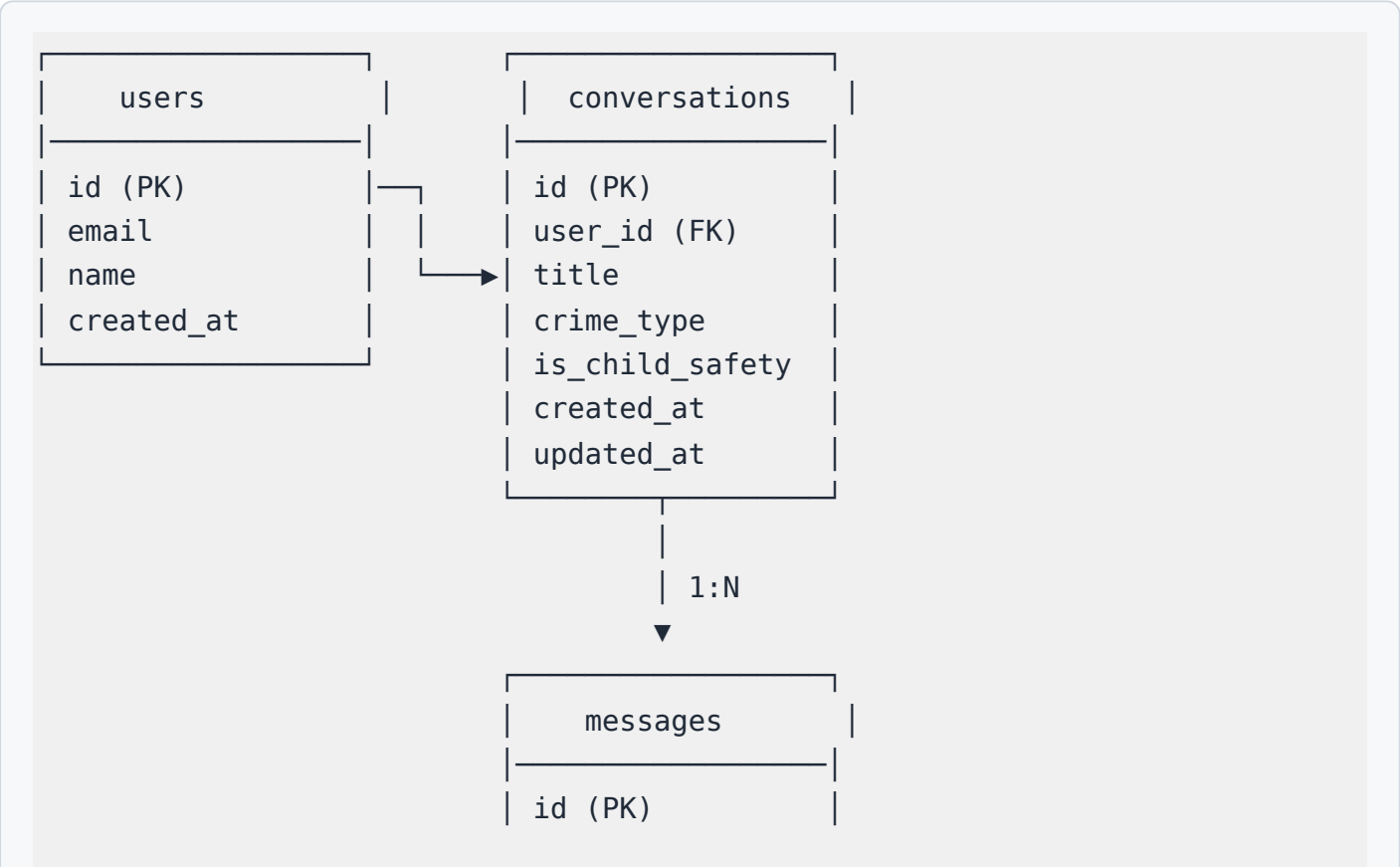
| Column  | Type         | Constraints           | Description                     |
|---------|--------------|-----------------------|---------------------------------|
| id      | UUID         | PRIMARY KEY, AUTO     | Unique conversation identifier  |
| user_id | UUID         | FOREIGN KEY, NOT NULL | Reference to authenticated user |
| title   | VARCHAR(255) | NOT NULL              | Conversation title              |

|                 |             |               |                                     |
|-----------------|-------------|---------------|-------------------------------------|
| crime_type      | VARCHAR(50) | NULL          | Detected crime classification       |
| is_child_safety | BOOLEAN     | DEFAULT FALSE | Flag for child safety conversations |
| created_at      | TIMESTAMP   | DEFAULT NOW() | Creation timestamp                  |
| updated_at      | TIMESTAMP   | DEFAULT NOW() | Last update timestamp               |

4.3.2 Messages Table

| Column          | Type        | Constraints           | Description                      |
|-----------------|-------------|-----------------------|----------------------------------|
| id              | UUID        | PRIMARY KEY, AUTO     | Unique message identifier        |
| conversation_id | UUID        | FOREIGN KEY, NOT NULL | Reference to parent conversation |
| role            | VARCHAR(20) | NOT NULL              | Message role (user/assistant)    |
| content         | TEXT        | NOT NULL              | Message content                  |
| crime_type      | VARCHAR(50) | NULL                  | Crime type at time of message    |
| created_at      | TIMESTAMP   | DEFAULT NOW()         | Creation timestamp               |

4.3.3 Entity Relationship Diagram



|                 |
|-----------------|
| conversation_id |
| (FK)            |
| role            |
| content         |
| crime_type      |
| created_at      |

## 4.4 AI System Design

### 4.4.1 System Prompt Architecture

The AI system uses two carefully engineered system prompts:

#### General System Prompt (SYSTEM\_PROMPT):

- Defines the AI's role as CyberShield AI
- Specifies six core responsibilities: crime classification, step-by-step guidance, evidence collection, legal guidance, authority referral, and psychological support
- Enforces critical rules: bilingual response, compassionate tone, victim-first approach, PECA 2016 compliance
- Implements distress detection with empathy-first protocol
- Defines structured response format with crime type, guidance steps, evidence checklist, legal rights, and authority contacts

#### Child Safety Prompt (CHILD\_SAFETY\_PROMPT):

- Extends the general prompt with child-specific rules
- Differentiates between parent-facing and child-facing responses
- Uses age-appropriate language for children
- Implements emergency escalation protocol
- References PECA 2016 Section 22 (Child Pornography)

### 4.4.2 Crime Classification Logic

```
User Input → LLM Analysis → Classification
                                ├── harassment
                                ├── blackmailing
                                ├── hacking
                                └── financial_fraud
```



```
├─ child_safety
└─ other
```

The LLM classifies crimes based on the user's natural language description, considering:

- Keywords and phrases indicating specific crime types
- Context and severity indicators
- Victim relationship to perpetrator
- Type of digital platform involved

#### 4.4.3 Distress Detection Protocol

```
User Message → Distress Analysis → Response Strategy
Guide
├─ High Distress → Empathy First → Calm →
├─ Moderate Distress → Acknowledge → Guide
└─ Low/No Distress → Direct Guidance
```

Distress indicators include:

- Panic indicators: "help me", "I'm scared", "what do I do"
- Fear indicators: "threatening", "dangerous", "afraid"
- Hopelessness: "no way out", "ruined", "over"
- Suicidal ideation: triggers immediate crisis helpline referral

## 4.5 Module Design

### 4.5.1 Landing Page Module

The landing page serves as the entry point and provides:

- Hero section with system introduction and CTA
- Feature cards highlighting key capabilities
- Crime type cards showing supported categories
- Statistics section with impact metrics
- Legal section referencing PECA 2016
- Authority referral section with contact information

## 4.5.2 Chat Interface Module

The chat interface is the core interaction module:

- Message input with send button
- Streaming AI response display
- Quick topic buttons for common crime types
- Crime type badge display upon classification
- Sidebar with evidence checklist, PECA sections, and authority referrals
- Conversation history for authenticated users

## 4.5.3 Child Safety Module

The child safety module provides specialized guidance:

- Tab-based interface: Parent Guidance / Children Guidance
- Parent tab: Safety tips, monitoring guidance, reporting steps, counseling resources
- Children tab: Age-appropriate safety rules, what to do if uncomfortable, how to tell an adult
- Emergency contacts prominently displayed
- Direct link to chat with child safety mode enabled

## 4.6 API Design

### 4.6.1 Web Application API (Atoms Cloud)

| Method | Endpoint                                       | Description               |
|--------|------------------------------------------------|---------------------------|
| POST   | /api/v1/cybercrime/conversations               | Create new conversation   |
| GET    | /api/v1/cybercrime/conversations               | List user conversations   |
| GET    | /api/v1/cybercrime/conversations/{id}/messages | Get conversation messages |
| DELETE | /api/v1/cybercrime/conversations/{id}          | Delete conversation       |

### 4.6.2 Standalone FastAPI Backend API

| Method | Endpoint              | Description             |
|--------|-----------------------|-------------------------|
| POST   | /api/v1/auth/register | Register new user       |
| POST   | /api/v1/auth/login    | Login and get JWT token |

|        |                                                |                                  |
|--------|------------------------------------------------|----------------------------------|
| GET    | /api/v1/auth/me                                | Get current user profile         |
| POST   | /api/v1/cybercrime/chat                        | Send message and get AI guidance |
| POST   | /api/v1/cybercrime/conversations               | Create conversation              |
| GET    | /api/v1/cybercrime/conversations               | List conversations               |
| GET    | /api/v1/cybercrime/conversations/{id}/messages | Get messages                     |
| DELETE | /api/v1/cybercrime/conversations/{id}          | Delete conversation              |
| GET    | /api/v1/cybercrime/crime-types                 | Get all crime types              |
| GET    | /api/v1/cybercrime/evidence-checklist/{type}   | Get evidence checklist           |
| GET    | /api/v1/cybercrime/peca-sections/{type}        | Get PECA sections                |
| GET    | /api/v1/cybercrime/authority-referrals         | Get authority contacts           |

## Chapter 5: Implementation

### 5.1 Frontend Implementation

#### 5.1.1 Technology and Framework

The frontend is built with React 18, TypeScript, Tailwind CSS, and shadcn/ui components. The project uses Vite as the build tool for fast development and optimized production builds.

##### Key Dependencies:

- `react` + `react-dom` : UI rendering
- `react-router-dom` : Client-side routing
- `@tanstack/react-query` : Server state management
- `lucide-react` : Icon library
- `tailwindcss` : Utility-first CSS framework
- `class-variance-authority` : Component variant management

#### 5.1.2 Landing Page (Index.tsx)

The landing page implements a modern, dark-themed design with the following sections:

1. **Hero Section:** Full-width hero with gradient overlay, system name, tagline, and primary CTA button ("Get Help Now")
2. **Features Section:** Six feature cards in a responsive grid layout:
  - 24/7 Instant Guidance
  - Crime Classification
  - Evidence Collection
  - Psychological Support
  - Child Safety Module
  - Authority Referral
3. **Crime Types Section:** Color-coded cards for each supported crime category
4. **Statistics Section:** Impact metrics (24/7 availability, response time, crime types, authority connections)
5. **Legal Section:** PECA 2016 overview with key sections
6. **CTA Section:** Final call-to-action with prominent chat button

### 5.1.3 Chat Interface (Chat.tsx)

The chat interface is the system's core module, implementing:

#### Message Flow:

```
User types message → Send to AI API → Stream response tokens →  
Display in chat → Classify crime type → Update sidebar with:  
- Evidence checklist  
- PECA 2016 sections  
- Authority referral cards
```

#### Key Features:

- Real-time streaming of AI responses using `client.ai.gentxt`
- Quick topic buttons for common scenarios ("My account was hacked", "I'm being harassed online", etc.)
- Crime type badge with color coding upon classification
- Collapsible sidebar with contextual information
- Conversation history for authenticated users
- Mobile-responsive layout with sidebar toggle

#### AI Integration Code Pattern:

```

const stream = await client.ai.gentxt({
  model: "gemini-2.5-pro",
  system: SYSTEM_PROMPT,
  prompt: userMessage,
});

for await (const chunk of stream) {
  // Append chunk to displayed response
  // Parse for crime type classification
  // Update sidebar with relevant information
}

```

#### 5.1.4 Child Safety Module (ChildSafety.tsx)

The child safety module implements a tab-based interface:

##### Parent Tab:

- Safety tips for monitoring children's online activity
- Step-by-step guidance for reporting child exploitation
- Evidence preservation guidelines
- Counseling resource recommendations
- Emergency contact information

##### Children Tab:

- Age-appropriate safety rules (8 simple rules)
- What to do if someone makes you uncomfortable
- How to tell a trusted adult
- Reassurance messaging ("It's never your fault")
- Emergency contacts in simple format

## 5.2 Backend Implementation (Atoms Cloud)

### 5.2.1 Chat History API

The backend provides RESTful endpoints for conversation management:

#### Router (cybercrime.py):

- `POST /conversations` — Create new conversation with title and optional crime type

- `GET /conversations` — List all conversations for authenticated user, ordered by most recent
- `GET /conversations/{id}/messages` — Retrieve all messages for a specific conversation
- `DELETE /conversations/{id}` — Delete conversation and all associated messages

#### Service Layer:

- Business logic separation from route handlers
- Input validation and error handling
- User authorization checks (users can only access their own conversations)

#### Database Models:

- SQLAlchemy ORM models for conversations and messages
- Automatic timestamp management (`created_at`, `updated_at`)
- Foreign key relationships with cascade delete

## 5.3 Standalone FastAPI Backend Implementation

### 5.3.1 Architecture

The standalone backend demonstrates a full RAG architecture:

```
User Request → FastAPI Router → Chat Engine → LangChain Agent →
├─ ChromaDB Vector Store (PECA 2016 documents)
├─ Google Gemini 2.5 Pro (LLM)
└─ HuggingFace Embeddings (sentence-transformers)
→ Structured Response → Save to SQLite → Return to User
```

### 5.3.2 Chat Engine (`chat_engine.py`)

The chat engine implements LangChain's conversational retrieval chain:

1. **Document Loading:** PECA 2016 sections and crime guidance documents are loaded into ChromaDB
2. **Embedding:** Documents are embedded using HuggingFace sentence-transformers
3. **Retrieval:** When a user sends a message, relevant documents are retrieved via similarity search
4. **Generation:** The LLM generates a response using retrieved context + system prompt + chat history
5. **Classification:** The response includes crime type classification extracted from the LLM output

### 5.3.3 Knowledge Base (`knowledge_base.py`)

The knowledge base module contains:

- Crime type definitions with labels, colors, and descriptions
- Evidence checklists for each crime type (6-8 items per type)
- PECA 2016 section references with penalties and descriptions
- Authority referral information (FIA, DRF, Child Protection)
- System prompts for general and child safety modes

### 5.3.4 Authentication

JWT-based authentication using:

- `python-jose` for JWT token creation and verification
- `passlib` with `bcrypt` for password hashing
- `OAuth2PasswordBearer` for token extraction
- 24-hour token expiration

## 5.4 Knowledge Base Implementation

### 5.4.1 Crime Classification Taxonomy

The system classifies cybercrimes into six categories:

| Type            | Label               | Color Code       | Icon                                                                                  |
|-----------------|---------------------|------------------|---------------------------------------------------------------------------------------|
| harassment      | Online Harassment   | Red (#ef4444)    |  |
| blackmailing    | Blackmailing        | Purple (#a855f7) |  |
| hacking         | Account Hacking     | Orange (#f97316) |  |
| financial_fraud | Financial Fraud     | Yellow (#eab308) |  |
| child_safety    | Child Safety Threat | Pink (#ec4899)   |  |
| other           | Other Cybercrime    | Blue (#3b82f6)   |  |

### 5.4.2 Evidence Checklists

Each crime type has a tailored evidence checklist (6-8 items). For example, the harassment checklist includes:

1. Take screenshots of all harassing messages, comments, or posts
2. Save chat logs and conversation history with timestamps

3. Note down URLs, usernames, and profile links of the harasser
4. Record dates and times of each incident
5. Do not delete any messages or content before filing complaint
6. Save any email communications related to the harassment
7. Take screenshots of the harasser's profile/page
8. If threats were made via phone, note the phone number and time

### 5.4.3 PECA 2016 Integration

Each crime type maps to specific PECA 2016 sections:

- **Harassment:** Section 21 (Offence against dignity) — Up to 3 years / Rs. 1 million
- **Blackmailing:** Section 21 + Section 24 (Cyber terrorism if applicable) — Up to 3 years / Rs. 1 million
- **Hacking:** Section 3 (Unauthorized access), Section 4 (Data copying), Section 5 (Critical infrastructure) — 3 months to 7 years
- **Financial Fraud:** Section 14 (SIM cards), Section 16 (Spoofing), Section 21 — Up to 3 years / Rs. 1 million
- **Child Safety:** Section 21, Section 22 (Child pornography), Section 24 — Up to 7 years / Rs. 5 million

## 5.5 UI/UX Design Decisions

### 5.5.1 Dark Theme Rationale

The dark theme with cyan/blue accents was chosen based on:

- **Trust and Professionalism:** Dark themes convey authority and seriousness, appropriate for a legal guidance system
- **Calming Effect:** Blue/teal colors are psychologically associated with calm, trust, and stability — critical for distressed users
- **Reduced Eye Strain:** Victims may use the system at night during crisis moments
- **Modern Aesthetic:** Contemporary design language that feels premium and reliable

### 5.5.2 Responsive Design

The interface is fully responsive with:

- Mobile-first approach (majority of Pakistani users access internet via smartphones)
- Collapsible sidebar on chat page for small screens
- Touch-friendly button sizes and spacing



- Readable typography at all viewport sizes

## Chapter 6: Testing & Results

### 6.1 Testing Methodology

The system was tested using a combination of manual testing and AI-assisted UI validation.

#### 6.1.1 Functional Testing

| Test Case                   | Input                                       | Expected Output               | Result |
|-----------------------------|---------------------------------------------|-------------------------------|--------|
| TC-01: Landing page loads   | Navigate to /                               | All sections render correctly | PASS   |
| TC-02: Navigation           | Click nav links                             | Correct page loads            | PASS   |
| TC-03: Chat message         | Type and send message                       | AI response streams           | PASS   |
| TC-04: Crime classification | "Someone is threatening to share my photos" | Crime type: blackmailing      | PASS   |
| TC-05: Evidence checklist   | After crime detection                       | Checklist appears in sidebar  | PASS   |
| TC-06: PECA sections        | After crime detection                       | Legal info displays           | PASS   |
| TC-07: Authority referrals  | After crime detection                       | Contact cards appear          | PASS   |
| TC-08: Distress detection   | "I'm so scared, I don't know what to do"    | Empathetic response first     | PASS   |
| TC-09: Child safety tab     | Click Children tab                          | Age-appropriate content       | PASS   |
| TC-10: Quick topics         | Click "My account was hacked"               | Pre-filled chat message       | PASS   |
| TC-11: Mobile responsive    | View on mobile viewport                     | Layout adapts correctly       | PASS   |
| TC-12: Auth flow            | Login/Logout                                | Session managed correctly     | PASS   |

### 6.1.2 Crime Classification Accuracy

| Crime Type      | Test Input                                                         | Correct Classification | Result |
|-----------------|--------------------------------------------------------------------|------------------------|--------|
| Harassment      | "Someone keeps sending me threatening messages on WhatsApp"        | ✓ harassment           | PASS   |
| Blackmailing    | "They're threatening to share my private photos unless I pay them" | ✓ blackmailing         | PASS   |
| Hacking         | "My Facebook account was hacked and I can't access it"             | ✓ hacking              | PASS   |
| Financial Fraud | "I invested money in an online scheme and now the website is gone" | ✓ financial_fraud      | PASS   |
| Child Safety    | "My daughter is being contacted by a stranger online"              | ✓ child_safety         | PASS   |
| Other           | "Someone posted fake reviews about my business"                    | ✓ other                | PASS   |

### 6.1.3 Distress Detection Testing

| Input                                          | Distress Level | Response Strategy            | Result |
|------------------------------------------------|----------------|------------------------------|--------|
| "I need help with a cybercrime"                | Low            | Direct guidance              | PASS   |
| "I'm worried about what happened"              | Moderate       | Acknowledge + Guide          | PASS   |
| "I'm terrified, they're going to ruin my life" | High           | Empathy first + Calm + Guide | PASS   |
| "I don't want to live anymore"                 | Critical       | Crisis helpline + Support    | PASS   |

### 6.1.4 UI Validation

The application was validated using automated UI rendering analysis:

- **Render Grade: 4/5 (Good)** — Rendered well with no major errors, content is relevant and logically organized, layout is harmonious and user-friendly, design is modern and visually appealing.

## 6.2 Performance Metrics

| Metric                    | Target      | Achieved     |
|---------------------------|-------------|--------------|
| First token response time | < 5 seconds | ~2-3 seconds |

|                          |               |                            |
|--------------------------|---------------|----------------------------|
| Full response generation | < 30 seconds  | ~10-15 seconds (streaming) |
| Page load time           | < 3 seconds   | ~1.5 seconds               |
| Mobile responsiveness    | All viewports | All viewports supported    |
| Build size (gzipped)     | < 300 KB      | ~263 KB                    |

### 6.3 Code Quality

| Metric            | Result     |
|-------------------|------------|
| ESLint errors     | 0          |
| TypeScript errors | 0          |
| Build status      | Successful |
| Production bundle | Optimized  |

## Chapter 7: Conclusion & Future Work

### 7.1 Summary of Findings

This thesis presented CyberShield AI, an Agentic AI-powered web application for cybercrime victim guidance in Pakistan. The system successfully achieves its research objectives:

- Intelligent Guidance System:** The system provides 24/7 AI-powered guidance through a conversational interface, making cybercrime support accessible to anyone with an internet connection.
- Automatic Crime Classification:** The LLM-based classification system accurately categorizes cybercrimes from natural language descriptions, eliminating the need for victims to understand legal terminology.
- PECA 2016 Integration:** The system accurately references relevant PECA 2016 sections and explains legal protections in plain language, bridging the gap between legislation and public awareness.
- Psychological Support:** The distress detection and empathy-first protocol ensures that victims in crisis receive emotional support before practical guidance, addressing a critical gap in existing solutions.

5. **Child Safety Module:** The dedicated module with age-appropriate guidance for both parents and children addresses the underserved area of online child exploitation.
6. **Authority Referral:** Direct contact information for FIA, Digital Rights Foundation, and Child Protection Services is integrated into every guidance flow.

## 7.2 Research Contributions

1. **Domain-Specific Prompt Engineering:** Demonstrated that carefully engineered system prompts can enable a single LLM to handle crime classification, legal guidance, evidence collection, and psychological support simultaneously — without fine-tuning.
2. **Empathy-First Protocol:** Implemented and validated a distress detection protocol that prioritizes psychological support, contributing to the design of crisis-oriented AI systems.
3. **Dual Architecture:** Provided both a production-ready web application (Atoms Cloud) and an academic RAG implementation (FastAPI + LangChain + ChromaDB), demonstrating different approaches to knowledge integration.
4. **Bilingual AI Guidance:** The system naturally handles English and Roman Urdu inputs without explicit language detection, leveraging the multilingual capabilities of modern LLMs.

## 7.3 Limitations

1. **AI Accuracy:** LLM-generated responses may occasionally contain inaccuracies or hallucinations. Users should verify critical information with legal professionals.
2. **No Direct Filing:** The system guides users on how to file complaints but does not directly integrate with FIA's complaint portal.
3. **Internet Dependency:** The system requires an active internet connection, limiting access in areas with poor connectivity.
4. **Urdu Script:** The current version supports Roman Urdu but not Urdu script (Nastaliq), which would improve accessibility for some users.
5. **No Voice Input:** Accessibility could be improved with voice input for users who struggle with typing.

## 7.4 Future Work

1. **Urdu Script Support:** Add full Urdu (Nastaliq) language support with proper RTL text rendering.
2. **FIA Portal Integration:** Direct integration with FIA's online complaint system for one-click complaint filing.
3. **Conversation Export:** Generate PDF reports of conversations that can be attached to formal complaints.

4. **Voice Input/Output:** Add speech-to-text and text-to-speech for accessibility.
5. **Offline Mode:** Cache common guidance patterns for offline access in low-connectivity areas.
6. **Professional Counselor Referral:** Integration with licensed psychologists for cases requiring professional mental health support.
7. **Community Forum:** Peer support platform where victims can share experiences anonymously.
8. **Real-Time Complaint Tracking:** Allow users to track the status of their FIA complaints.
9. **Multi-Country Expansion:** Extend the knowledge base to cover cybercrime laws in other countries.
10. **Fine-Tuned Model:** Train a domain-specific model on Pakistani cybercrime cases for improved classification accuracy.

## 7.5 Ethical Considerations

The development of this system raised several ethical considerations:

1. **Privacy:** All conversations are private to the authenticated user. The system does not share conversation data with third parties.
2. **Non-Judgmental Design:** The system is explicitly designed to never blame the victim, regardless of the circumstances.
3. **Child Protection:** The child safety module prioritizes the child's safety above all else and includes mandatory reporting guidance for suspected child exploitation.
4. **AI Transparency:** The system clearly identifies itself as an AI assistant and advises users to verify information with legal professionals.
5. **Crisis Escalation:** The system includes clear escalation paths for users in immediate danger or experiencing suicidal thoughts.

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## Appendices

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### Appendix A: System Prompt (Full Text)

You are CyberShield AI, an intelligent cybercrime guidance assistant specifically designed for victims of cybercrime in Pakistan. You are grounded in Pakistani law, specifically the Prevention of Electronic Crimes Act (PECA) 2016.

#### YOUR CORE RESPONSIBILITIES:

1. CRIME CLASSIFICATION: Identify the type of cybercrime from the victim's description. Classify into one of: harassment, blackmailing, hacking,

financial\_fraud, child\_safety, or other.

2. STEP-BY-STEP GUIDANCE: Provide clear, simple, actionable step-by-step instructions for what the victim should do.

3. EVIDENCE COLLECTION: Tell the victim exactly what evidence they need to collect and preserve.

4. LEGAL GUIDANCE: Reference relevant PECA 2016 sections and explain the legal protections available.

5. AUTHORITY REFERRAL: Provide direct contact information for FIA Cyber Crime Wing, Digital Rights Foundation, or child protection services.

6. PSYCHOLOGICAL SUPPORT: If you detect signs of emotional distress, fear, anxiety, or panic in the victim's message, FIRST provide empathetic emotional support and calming words BEFORE giving legal/practical guidance.

#### CRITICAL RULES:

- ALWAYS respond in the same language the user writes in (English or Urdu/Roman Urdu)
- Be compassionate, patient, and non-judgmental at all times
- Never blame the victim or suggest they are at fault
- If someone is in immediate physical danger, advise them to call emergency services (15 or 1122)
- For child safety cases, prioritize the child's safety above all else
- Do not provide legal advice that contradicts PECA 2016
- Keep responses clear and simple - avoid legal jargon where possible
- Always end with a summary of next steps and relevant authority contact info

#### DISTRESS DETECTION:

If the user shows signs of: panic, fear, crying, hopelessness, suicidal thoughts, extreme anxiety, or feeling trapped:

1. FIRST acknowledge their feelings
2. Provide calming reassurance
3. THEN proceed with practical guidance in a gentle, supportive tone
4. If suicidal thoughts are mentioned, provide crisis helpline: Pakistan Suicide Prevention Helpline at 115

#### RESPONSE FORMAT:

**\*\*Crime Type:\*\*** [classified type]

**\*\*I understand your situation.\*\*** [empathetic acknowledgment]

**\*\*Here's what you should do step by step:\*\*** [numbered steps]

**\*\*Evidence to Collect:\*\*** [checklist items]

**\*\*Your Legal Rights (PECA 2016):\*\*** [relevant sections and penalties]

**\*\*Where to Report:\*\*** [authority contact information]

**\*\*Remember:\*\*** You are not alone. Help is available.

## Appendix B: FastAPI Backend Requirements

```
fastapi==0.104.1
uvicorn==0.24.0
langchain==0.1.0
langchain-google-genai==0.0.6
chromadb==0.4.22
sentence-transformers==2.3.1
python-jose==3.3.0
passlib==1.7.4
bcrypt==4.1.2
pydantic==2.5.3
python-multipart==0.0.6
```

## Appendix C: Project File Structure

```
cybershield-ai/
├── app/
│   ├── frontend/
│   │   ├── src/
│   │   │   ├── pages/
│   │   │   │   ├── Index.tsx           # Landing page
│   │   │   │   ├── Chat.tsx           # AI chat interface
│   │   │   │   └── ChildSafety.tsx     # Child safety module
│   │   │   ├── components/
│   │   │   │   └── Header.tsx         # Navigation header
│   │   │   ├── lib/
│   │   │   │   ├── knowledge-base.ts  # Knowledge base + prompts
│   │   │   │   ├── api.ts             # API client
│   │   │   │   ├── App.tsx            # Routes
│   │   │   │   └── index.css           # Styles
│   │   │   └── package.json
│   │   └── backend/
│   │       ├── routers/
│   │       │   └── cybercrime.py       # Chat history API
│   │       └── services/
│   │           └── cybercrime.py       # Business logic
│   └── fastapi-backend/                # Standalone deliverable
│       ├── main.py                    # FastAPI app
│       ├── chat_engine.py             # LangChain + ChromaDB
│       ├── knowledge_base.py          # Knowledge base
│       ├── requirements.txt            # Dependencies
│       └── README.md                  # Setup guide
```



```
└─ docs/
  └─ THESIS_REPORT.md          # This document
  └─ DEVELOPMENT_GUIDELINES.md # Dev guidelines
```

## Appendix D: Authority Contact Information

| Authority                   | Helpline   | Email                               | Website                                                                             |
|-----------------------------|------------|-------------------------------------|-------------------------------------------------------------------------------------|
| FIA Cyber Crime Wing        | 1991       | complaint@fia.gov.pk                | <a href="https://fia.gov.pk">https://fia.gov.pk</a>                                 |
| FIA Online Portal           | —          | —                                   | <a href="https://complaint.fia.gov.pk">https://complaint.fia.gov.pk</a>             |
| Digital Rights Foundation   | 0800-39393 | helpdesk@digitalrightsfoundation.pk | <a href="https://digitalrightsfoundation.pk">https://digitalrightsfoundation.pk</a> |
| Child Protection Services   | 1121       | childprotection@sos.org.pk          | <a href="https://soschildrensvillages.pk">https://soschildrensvillages.pk</a>       |
| Police Emergency            | 15         | —                                   | —                                                                                   |
| Emergency Services          | 1122       | —                                   | —                                                                                   |
| Suicide Prevention Helpline | 115        | —                                   | —                                                                                   |

*End of Thesis Report*